Process Analysis

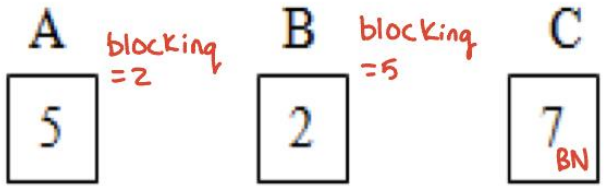
Inputs → Transformation → Outputs

Definitions	
Cycle Time	Average time between completions of successive units in a process (time/unit)
Throughput Time	Average time that it takes a unit to move through an entire process. Includes processing and blocking time (time)
Throughput Rate	Output rate that the process is expected to produce over a period of time (units/time)
Utilization	Ratio of time resource is used relative to that available (used/theoretical)
Bottleneck	A resource that limits the capacity of maximum output of the process. Can be when an employee works too slowly in a multi-stage process and their work piles up.
Pacing	Fixed timing of the movement of items through the process
Blocking	Occurs when activities in the stage must stop because there is no place to deposit the item just completed (**adds to throughput time**)
Starving	Occurs when the activities in a stage must stop because there is no work (**does not add to throughput time**)
Buffer	A storage area between stages where the output of a stage is placed prior to being used in a downstream stage (holding area).
Efficiency	A ratio of the actual output of a process relative to some standard
Pull - Make to order	System activated in response to an actual order. WIP and finished goods inventory is kept to a minimum
Push- Make to stock	Process or system activated to meet expected or forecast demand. Customer orders served from target stock levels
Process Flowcharting	Use of a diagram to present major elements of a process (tasks/operations, flow of materials/customers, decision point, storage areas/queues, etc). Often a first step in analyzing a process. Provides a good visual for those not familiar with the process. Helps identify waste!
Methods to Improve Process Performance	Perform activities in parallel, change the sequence of activities, reduce interruptions, etc…
Equations	
Little's Law *(assumes process is in steady state!)*	Book's definition: Inventory = Throughput rate X Flow Time $Throughput\ (time) = \frac{WIP\ (units)}{throughput\ rate\ (units/time)}$ $Throughput\ (time) = WIP\ (units) * \text{cycle time (time/unit)}$
Throughput Rate and Cycle Time	throughput rate = 1/cycle time → cycle time = 1/throughput rate
Efficiency	$Efficiency = \frac{\sum Processing\ Time}{N*CT}$ = Actual / Standard N = number of stations CT = cycle time
Velocity	$\text{Velocity} = \frac{\text{value added time}}{\text{throughput time}}$

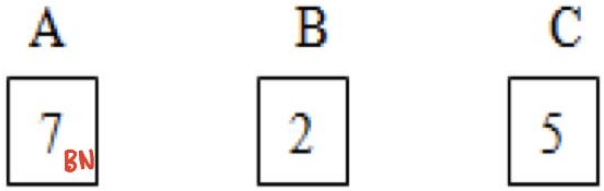
Problem 2

Fill in the table below based on the following work stations and processing times:

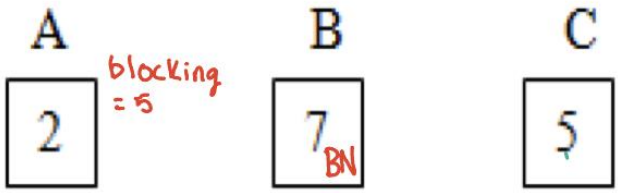
1)



2)



3)



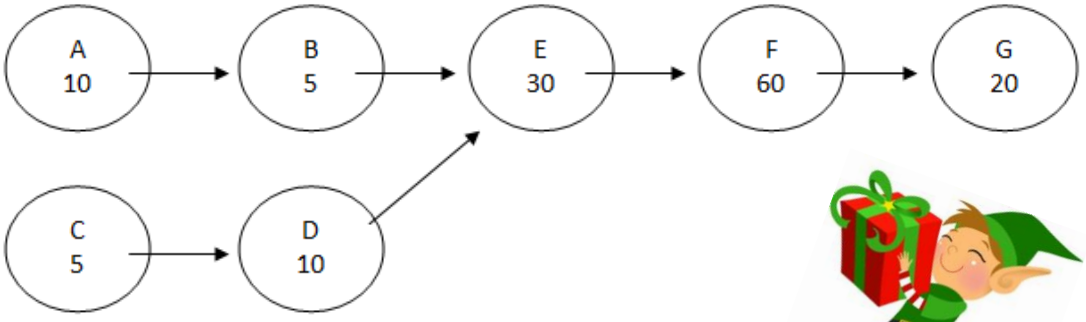
4)



Process	Bottleneck Operation	Activities that experience starving	Activities that experience blocking	Throughput time (process + blocking)	Cycle Time (determined by BN)	Efficiency ($\frac{\Sigma\, processing}{N*CT}$)
1	C	-	A and B	14 (process) + 7 (blocking) = 21	7	$\frac{14}{3 \times 7} \times 100 = 66.7\%$
2	A	B and C	-	14 (process) = 14	7	$\frac{14}{21} \times 100 = 66.7\%$
3	B	C	A	14 + 5 = 19	7	$\frac{14}{21} \times 100 = 66.7\%$
4	B and D	C	A	18 + 5 = 23	7	$\frac{18}{28} \times 100 = 64.3\%$

Problem 3

The assembly line depicts each operation (seconds) used to make a toy in Santa's workshop which runs for 8 hours each day. Work in process inventory is not allowed to accumulate between workstations and there is currently one worker elf at each operation.





a) What is the bottleneck of the process?

F – limits production by having the longest processing time

b) Which operations experience blocking and which experience starving?

Upstream operations are blocked – A,B,C,D,E
Downstream is starved - G

c) What is the cycle time for this assembly line?

60 sec/toy (determined by bottleneck)

d) If the workshop runs 8 hour a day and there are 5 days left until Christmas, what is the maximum number of this toy that can be produced?

$$5\ days\left(\frac{8hrs}{day}\right)\left(\frac{60min}{hr}\right)\left(\frac{60sec}{\text{min}}\right) = 144{,}000\ seconds\ available$$

$$\max output = \frac{144{,}000\ seconds}{60\ sec/toy} = \mathbf{2{,}400\ toys}$$

e) In steady state (no startup), what is the throughput time?
*Remember that this is composed of process time and blocking. Be careful with parallel operations.
*note A, B and C, D are in parallel, don't double up on their time!
*upstream BN causes previous steps to essentially take onto its pace

$$\text{So } [60(A\ or\ C) + 60(B\ or\ D) + 60(E)] + 60(F) + 20(G) = \mathbf{260\ seconds}$$

f) Santa is considering creating workstations (rather than one elf at each step). He still needs 800 of these toys (and we still operate for 8 hours over the next 5 days). Using what you know about line balancing, what would the required workstation cycle time have to be?

$$CT = \frac{time\ available}{\#\ units\ required} = \frac{144{,}000\ sec}{800\ units} = \mathbf{180}\frac{\boldsymbol{sec}}{\boldsymbol{toy}}$$